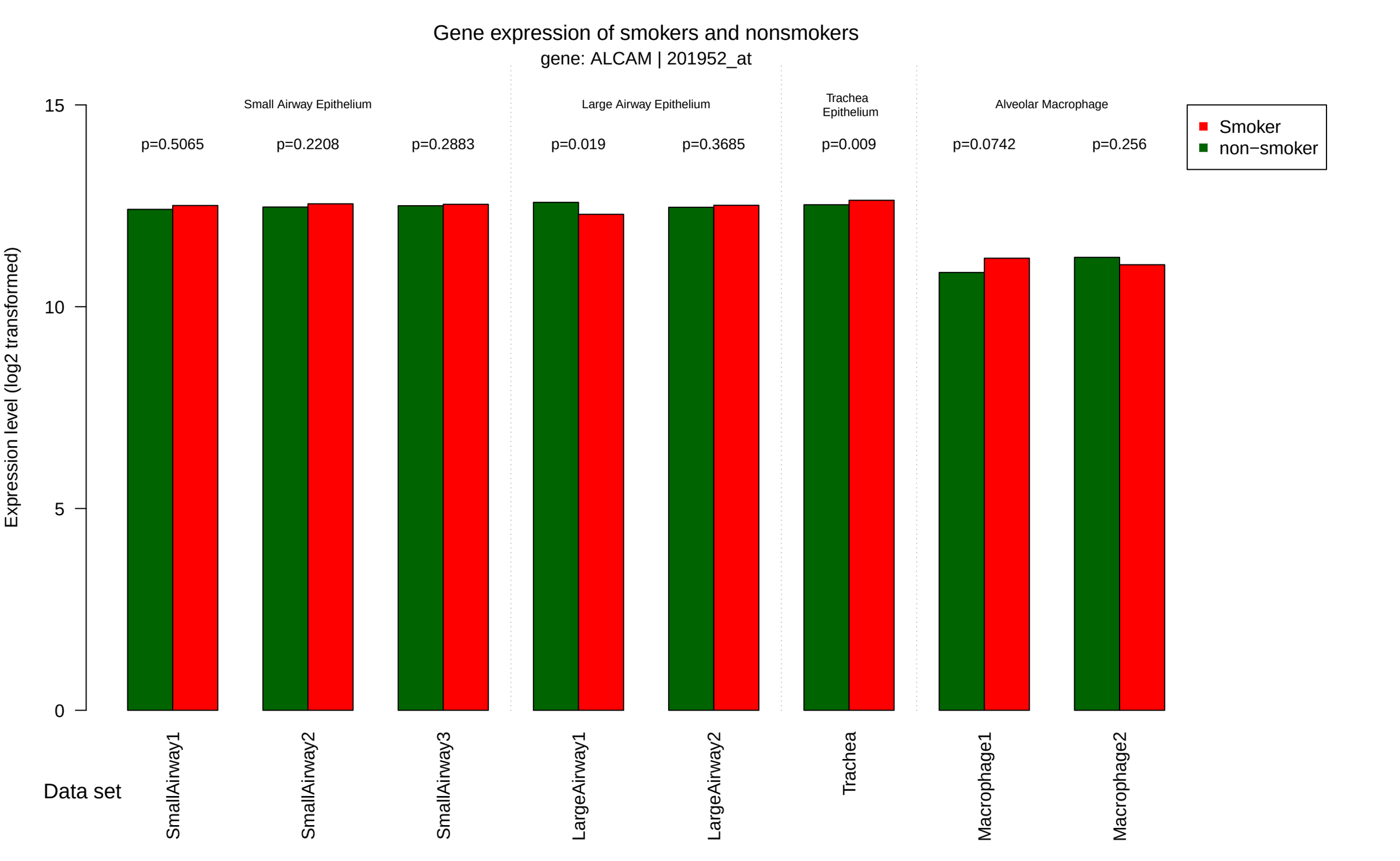
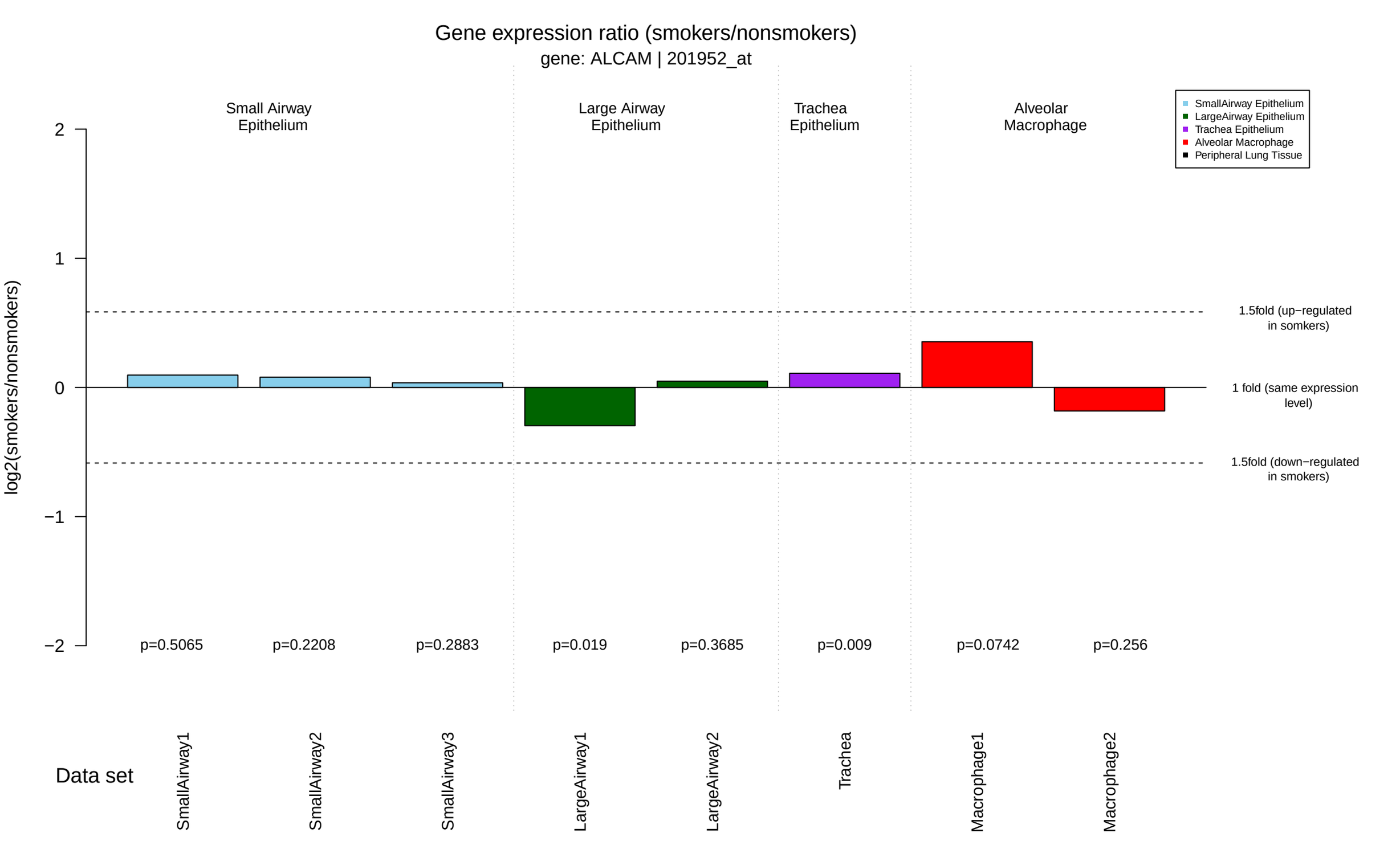
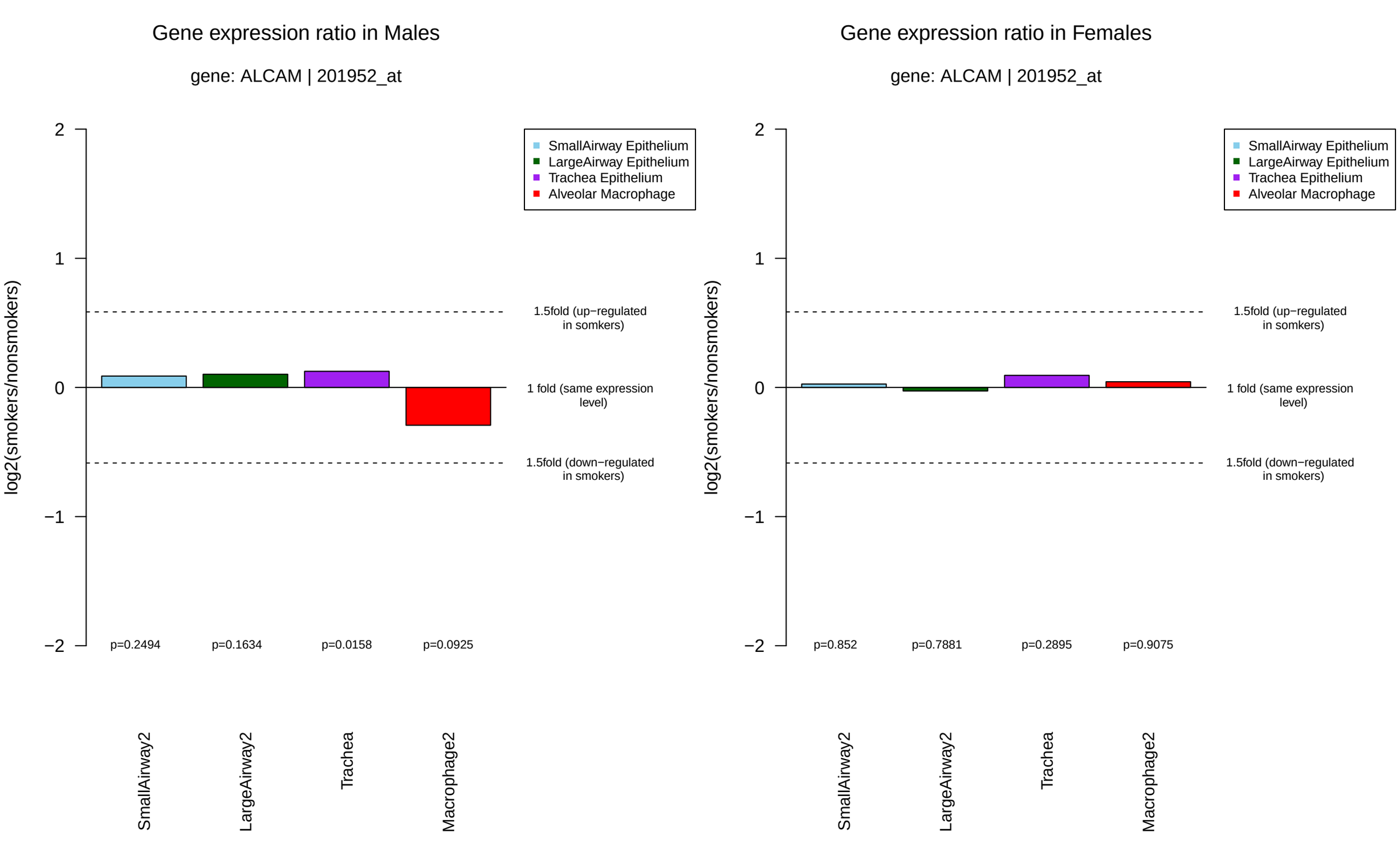
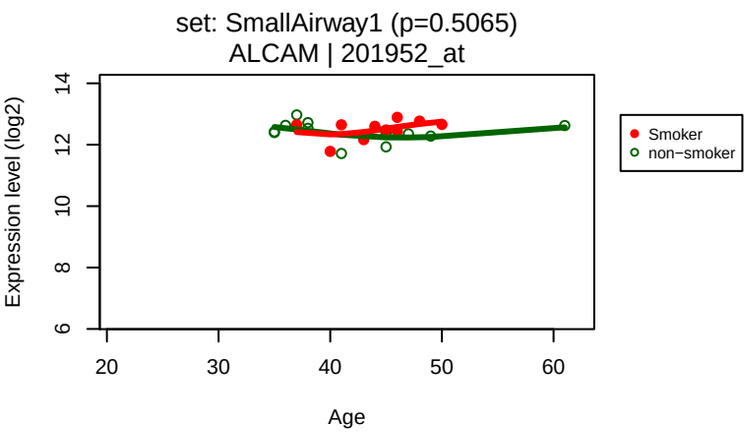
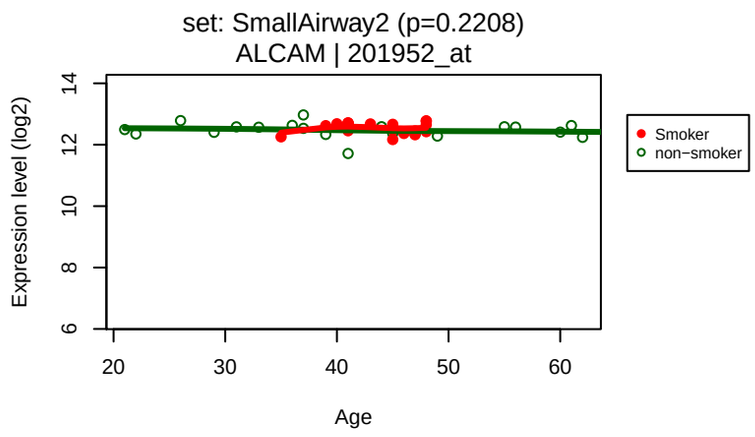
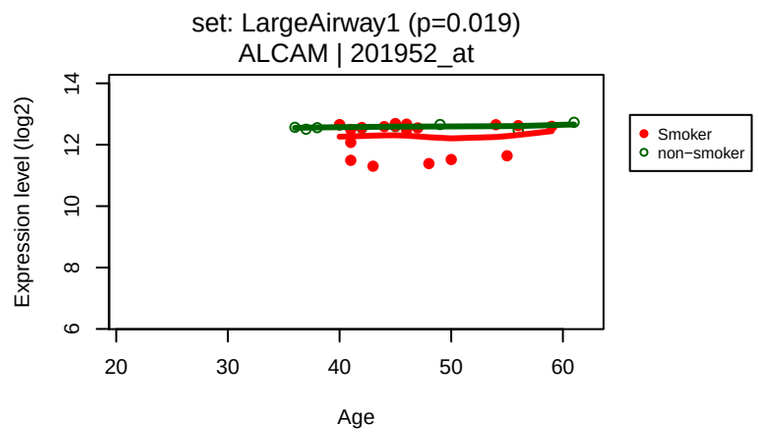

Gene expression of smokers and nonsmokers
gene: ALCAM | 201952_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5065
p=0.2208
p=0.2883
p=0.019
p=0.3685
p=0.009
p=0.0742
p=0.256
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

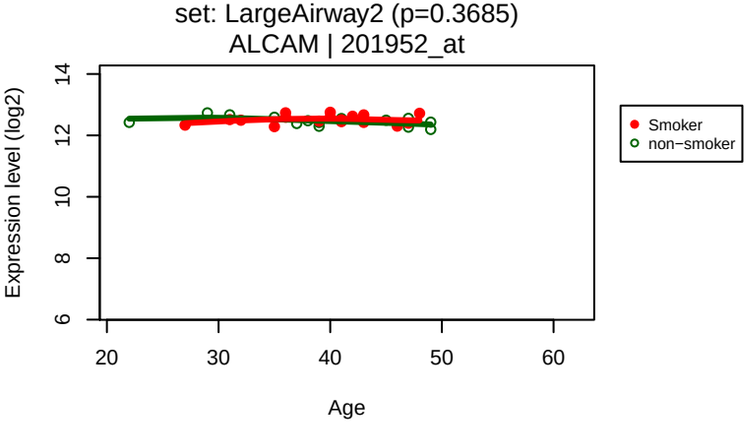

Gene expression ratio (smokers/nonsmokers)
gene: ALCAM | 201952_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5065
p=0.2208
p=0.2883
p=0.019
p=0.3685
p=0.009
p=0.0742
p=0.256
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

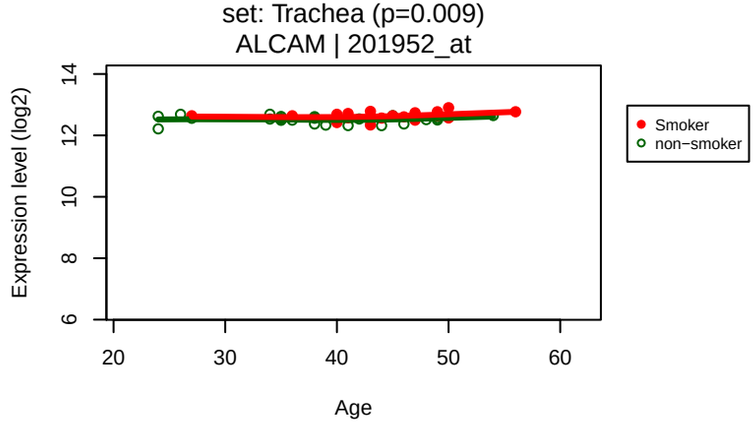

Gene expression ratio in Males
gene: ALCAM | 201952_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2494
p=0.1634
p=0.0158
p=0.0925
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALCAM | 201952_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.852
p=0.7881
p=0.2895
p=0.9075
SmallAirway2
LargeAirway2
Trachea
Macrophage2

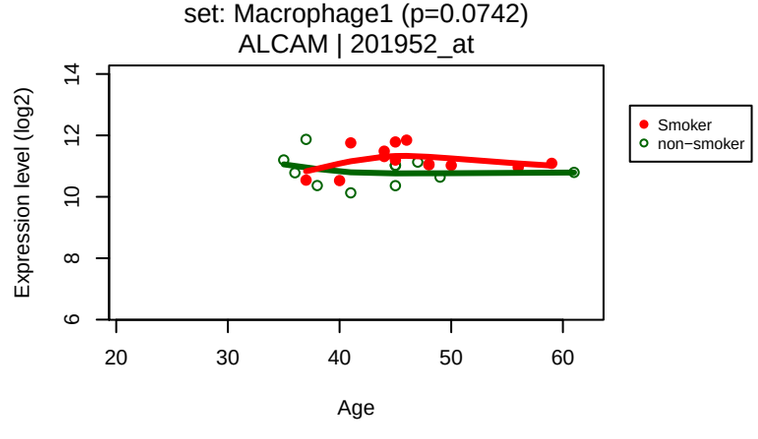

set: SmallAirway1 (p=0.5065)
ALCAM | 201952_at
Expression level (log2)
Age
Smoker
non−smoker

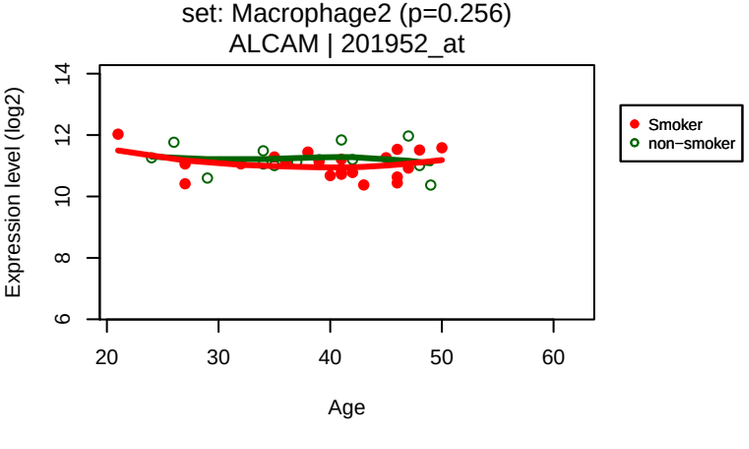

set: SmallAirway2 (p=0.2208)
ALCAM | 201952_at
Expression level (log2)
Age
Smoker
non−smoker

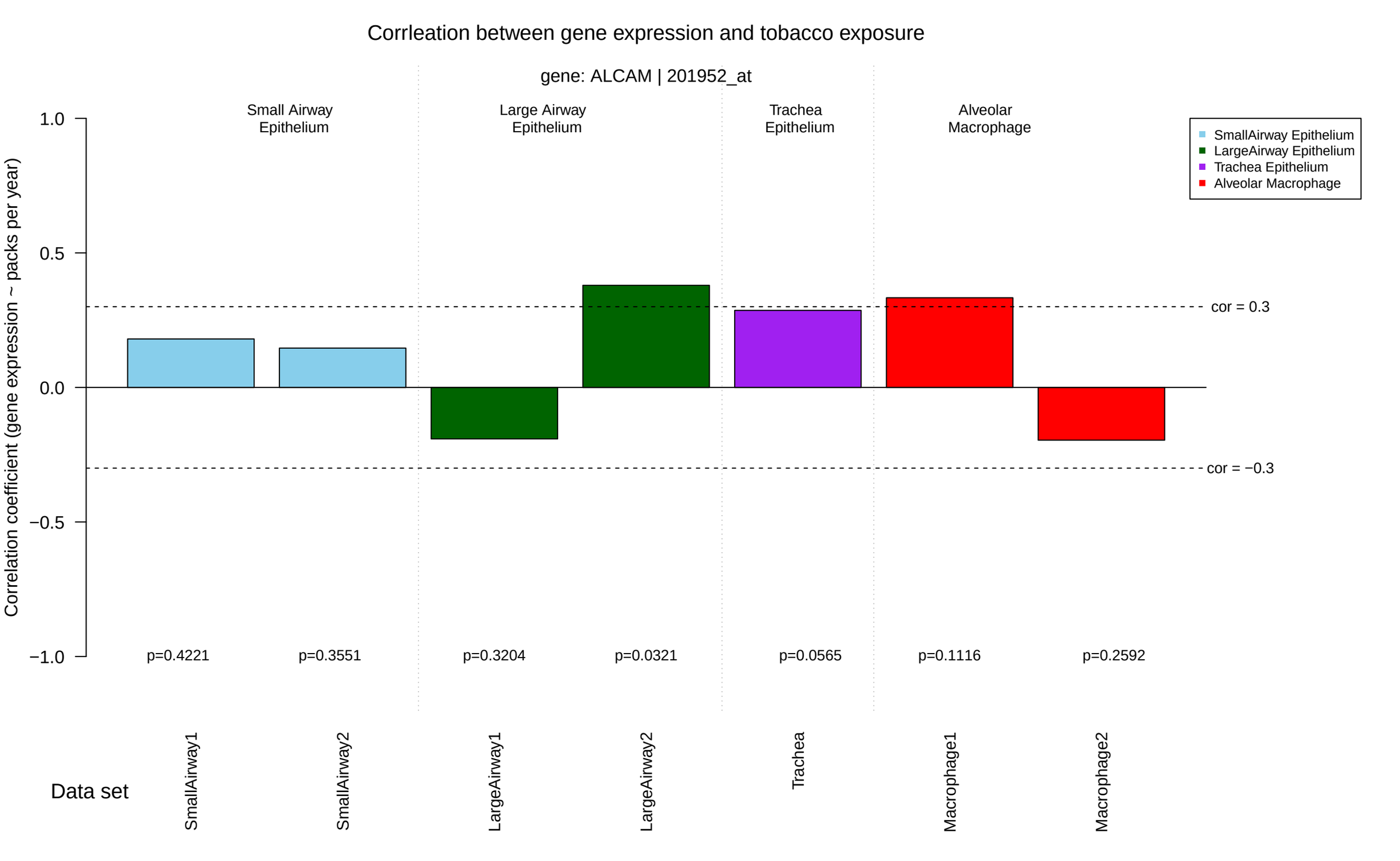

set: LargeAirway1 (p=0.019)
ALCAM | 201952_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3685)
ALCAM | 201952_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.009)
ALCAM | 201952_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0742)
ALCAM | 201952_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.256)
ALCAM | 201952_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALCAM | 201952_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4221
p=0.3551
p=0.3204
p=0.0321
p=0.0565
p=0.1116
p=0.2592
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2